CSC_51053_EP, Exercise Session: FDs and Concurrency Control

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The following comment applies to all exercises on FDs: *Most of the time, we work with dependencies having* ***a single attribute on the right-hand side****, because: this is the form in which 3NF is checked; and it is always possible to obtain such a cover from an arbitrary set of dependencies.*
We work with dependencies having several attributes on the right-hand side only when we need them for a specific axiom application step.

1 Keys, 3NF and BCNF [2]

Consider a relation $R(A, B, C, D, E)$ on which the following dependencies hold: $A \to B$, $BC \to E$, and $DE \to A$.

1. List all keys for R.
2. Is R in third normal form (3NF)?
3. Is R in Boyce-Codd Normal Form (BCNF)?

2 Decomposition into BCNF [2]

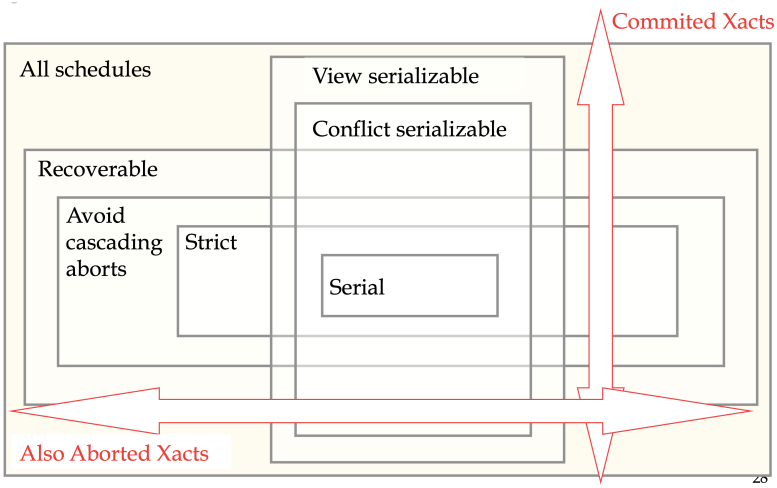
You are given a relation $R(A, B, C, D, E, G, H)$ and a set of functional dependencies $F = \{AB \to C, AC \to B, AD \to E, B \to D, BC \to A, E \to G\}$.

1. For each of the following attribute sets, do the following. (i) Find a minimal cover of the set of dependencies that hold over the relation. (ii) What is the strongest normal form (3NF or BCNF) that a relation with these attributes satisfies? (iii) If that relation is not in BCNF, decompose it into BCNF.
   (a) $CDEGH$
   (b) ABC
   (c) $ABCD$
   (d) $ABCEG$
   (e) $ACEH$
2. Only for the purpose of this point, assume the dependency $G \to H$ also holds. Redo the above analysis for $ACEH$.
3. Which of the following decompositions of $R(ABCDEG)$, with the same set of dependencies F, are: dependency-preserving? lossless-join?
   (a) $\{AB, BC, ABDE, EG\}$
   (b) $\{ABC, ACDE, ADG\}$

3 Schedule properties

3.1 Classes of Schedules [2]

Consider the following classes of schedules: *serializable, conflict-serializable, view-serializable, recoverable, avoids-cascading-aborts,* and *strict*. For each of the following schedules, state which of the preceding classes it belongs to (recall the inclusion diagram below):



The actions are listed in the order they are scheduled and prefixed with the transaction name.

1. T1:R(X), T2:R(X), T1:W(X), T2:W(X), T2:Commit, T1: Commit
2. T1:W(X), T2:R(Y), T1:R(Y), T2:R(X), T1: Commit, T2: Commit
3. T1:R(X), T2:R(Y), T3:W(X), T2:R(X), T1:R(Y): T1:Commit, T2:Commit, T3:Commit
4. T1:R(X), T2:R(Y), T3:W(X), T2:R(X), T1:R(Y): T1:Commit, T3:Commit, T2:Commit
5. T1:R(X), T2:W(X), T1:W(X), T2:Commit, T1:Commit
6. T1:R(X), T2:W(X), T1:W(Y), T2:R(Y), T2:Commit, T1:Commit

3.2 More on Schedules [1]

Consider the schedule S shown below (operations are shown in temporal order, from left to right; we number the writes and reads to facilitate refering to them):

T_1			$R_1(A)$		$W_1(B)$				
T_2	$R_2(B)$	$W_2(A)$				$W_2(B)$			
T_3				$R_3(A)$			$W_3(B)$		

Answer the questions below providing a brief justification for your answer to each of them.

1. Is this schedule view serializable? If it is, propose a view-equivalent serial schedule.
2. Is this schedule conflict serializable? If it is, propose a conflict-equivalent serial schedule.
   Now consider the schedule S_2 where we also show transaction commits:

T_1			$R_1(A)$		$W_1(B)$						C_1	
T_2	$R_2(B)$	$W_2(A)$					$W_2(B)$			C_2		
T_3				$R_3(A)$				$W_3(B)$				C_3

3. Is S_2 recoverable?
4. Does S_2 avoid cascading aborts?
5. Is S_2 strict?

References

[1] Hector Garcia-Molina, Jeffrey D. Ullman, and Jennifer Widom. *Database systems - the complete book (2. ed.)*. Pearson Education, 2009.

[2] Raghu Ramakrishnan and Johannes Gehrke. *Database management systems (3rd ed.)*. McGraw-Hill, 2003.